

AIRSTREET · CALCULATION

AirStreet sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-30	Amish Chadha	Design made constructable (AST-DDR-003); mass, area, mounting loads and cost recalculated; R13 and R15 now not met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (power and energy, airtime and storage, NO2 error budget, radiation shield, PM2.5 humidity error, geometry, mass and area, wind and mounting, service, cost)

AirStreet sizing calculations

Version 0.3 follows the constructable design of AST-DDR-003 (2026-09-30), which made every part of the node buildable and moved onto FieldNode's own constructable core. On paper AirStreet now meets eight of its sixteen requirements (four by calculation, four by design), has five at risk and misses two; R3 stays withdrawn. The node weighs 3.84 kg on the pole against R13's 3.5 kg, and the sensor head costs \$303.00 against the \$285 `budget_usd`, so R13 and R15 are not met; both are proposed to Amish in AST-DDR-003 (A1, A2). The frontal area, 0.143 m², stays inside 0.15 m², and every new mounting part carries its load with a large margin. Power, airtime, the NO₂ error budget, the shield and the humidity results are unchanged from v0.2: the 60 s particle run gives an 87.0 mW design sensor load inside FieldNode's 100 mW allowance and 6.6 days without sun, and an annual NO₂ mean carries about 4.2 µg/m³ of uncertainty (1σ). Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that the node is safe to mount on any particular pole, and they are not a substitute for the pole owner's structural check, for electrical checks on the FieldNode cell or for field collocation of the sensors. AirStreet data are indicative, not a regulatory measurement or health advice. See AST-PRC-001, Safety.

■ Scope and method

The note checks every requirement in AST-REQ-001 v0.5 against the design in AST-PRC-001 v0.5 and the parametric model `cad/src/model.py`, which builds the FieldNode core from FieldNode's own model (vendored as `cad/src/fieldnode_core.py`). The script imports the model's `PARAMS`, derived dimensions and part solids, so the pod, shield, rail, clamp positions and projected areas used here are those in the STEP files and in drawing AST-DWG-001. It reads `bom/bom.csv` and `budget_usd` in `project.yaml`. FieldNode figures are taken from FND-CAL-001 (v0.1 for power and energy, v0.3 for mass) in the FieldNode repo. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`.

The design case is a 140 mm street light pole with the inlet plane 3.0 m above the sidewalk (AST-DDR-001 D7), one record every 5 min with a 60 s PM run (D8 and DDR-002), LoRaWAN at SF9 in EU868 through a private gateway (15 min on The Things Network), FieldNode's back plate left off (DDR-002), and ambient -10 to 50 °C.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
PM sensor	SPS30 at 5 V, 55 mA typical and 65 mA maximum in measurement mode; 41 x 41 x 12 mm, 26.3 g; up to 30 s to first reading at low particle counts; recommended 10 to 40 °C and 20 to 80 % RH	Sensirion SPS30 datasheet (07/2023), checked 2026-09-25
NO2 chain	Front end and ADC held on continuously, 1.5 mA at 5 V; overall gain 0.25 mV/ppb; 5 ppb (1 σ) noise per 1 s sample; 10 samples per record	Assumed; the Alphasense pages could not be fetched, so these are not checked
T and RH	SHT45, ± 0.1 °C and ± 1 % RH typical	Adafruit SHT45 product page, checked 2026-09-25
Margin	1.2 on the port load, applied to the maximum SPS30 current	Judgment
FieldNode core	Rails 90 % efficient; SX1262 +14 dBm at 45 mA, receive 4.6 mA, two 0.1 s windows, controller 0.5 s at 8 mA per report, 33 μ A sleep; 6 Ah LiFePO4 cell, 80 % usable; capacity ratios 0.70 at -20 °C and 0.80 at end of life; 7.75 Wh/day stored in the worst month, 0.8 Wh/day on a hot clear day without a sun shield; 2.41 kg; \$126.00	FND-CAL-001 v0.1
Extra controller time	1 s at 8 mA per record to run and read the sensors	Assumed
NO2 calibration	Field-calibrated residual MAE 3.5 ppb, taken as $\sigma = 4.39$ ppb; zero drift 0.5 ppb/month, half removed by the anchor node; 2 ppb bias moving from the reference site to a street; 6 h correlation time	Residual from Zimmerman et al., 2018 ; the rest assumed
PM humidity growth	κ -Köhler volume growth $1 + \kappa \cdot a_w / (1 - a_w)$ with $\kappa = 0.3 \pm 0.1$; residual after the CalRig transfer check $2 \mu\text{g}/\text{m}^3 + 5\%$	Assumed typical urban aerosol
Shield	1,000 W/m ² sun, ground albedo 0.2, absorptance 0.25 (aged white ASA), 15 W/m ² K per plate face, view factors from the probe of 0.1 (cap), 0.1 (bottom plate) and 0.6 (other plates, 0.5 K warm); inner air speed 10 to 30 % of a 1 m/s wind	Handbook ranges; assumed
Mass	ASA 1.07, aluminum 2.70, stainless 7.90 g/cm ³ from model volumes; FieldNode V-blocks and small clamps (0.20 kg) and back plate (0.47 kg) left off	Estimates
Wind	35 m/s gust; air 1.225 kg/m ³ ; drag coefficient 1.2 on the projected area; band preload 1,000 N (as FND-CAL-001), friction 0.3	Screening values, not a code check

■ A. Power and energy (R9)

- **PM sensor.** 0.275 W while running at 20 % duty (60 s in 300 s) is 55.0 mW typical, or 65.0 mW at the datasheet maximum [A1].
- **Port load.** The NO2 front end and ADC add 7.5 mW continuously and the T and RH sensor 0.01 mW, for 62.5 mW typical [A2]. With the maximum SPS30 current and the 1.2 margin, the design load is 87.0 mW, against FieldNode's 100 mW design value and 115 mW ceiling [A3]. R9 is met on paper under either figure, with 13 mW left under the design value.

- **Core.** At a 5 min interval each uplink costs 52.9 mJ at SF9, and the core uses 9.0 mWh/day including sensor wake-ups and sleep, against 4.0 mWh/day for FieldNode alone at 15 min [A4].
- **Daily draw.** 2.33 Wh/day from the cell, a 97.1 mW average [A5].
- **Autonomy.** A full cell lasts 6.6 days without sun, 4.6 days at -20 °C and 5.3 days at end of cell life [A6]. The cold figure is below the 5 days FieldNode's own R6 asks of its core; AirStreet's R9 does not set an autonomy target.
- **Energy balance.** In FieldNode's worst month the cell stores 7.75 Wh/day against 2.33 Wh drawn, 3.3 times the demand [A7]. On a hot clear day without FieldNode's sun shield only 0.8 Wh is stored, a 1.53 Wh/day deficit, which a full cell covers for 10 such days in a row [A8]; the shield is required above 45 °C (DDR-002).
- **Clean air.** The SPS30 needs up to 30 s to its first reading at low particle counts, so the 60 s run (DDR-002) leaves at least 30 s of valid readings. The 30 s run of v0.1 gave 48.0 mW [A9].

■ B. Data, airtime and storage (R8, R10)

- **Record.** Eight 2-byte values (PM1, PM2.5, PM10, WE, AE, T, RH, status), a 2-byte counter and 2 bytes of flags: 20 bytes, 288 records a day [B1].
- **Airtime.** With 13 bytes of LoRaWAN overhead on a 125 kHz channel, one uplink takes 72 ms at SF7, 134 ms at SF8, 247 ms at SF9, 453 ms at SF10 and 1,810 ms at SF12 [B2]. The TRL 2 figure of about 1.3 s at SF12 was low.
- **EU868 duty cycle.** The worst case, SF12, uses 0.60 % of each 5 min, so the 1 % sub-band limit is met at every spreading factor [B3]. R10 is met on paper.
- **Fair use.** On The Things Network's 30 s/day fair-use policy, a 5 min interval fits only at SF7 (20.7 s); SF8 needs 38.5 s and SF9 71.1 s [B4]. Under DDR-002 the node reports every 5 min only through a private gateway (TwinKit by default, FND-DDR-001 D7), where this limit does not apply, and every 15 min on TTN. At 15 min an uplink takes 12.8 s/day at SF8 and 23.7 s/day at SF9, inside 30 s, but 43.5 s/day at SF10, so fair use is met up to SF9 [B8].
- **US915.** The 400 ms dwell limit allows SF9 (247 ms) and not SF10 (453 ms) [B5].
- **Storage.** 7 days of store and forward take 40.3 kB of FieldNode's 16 MB flash [B6], and an hourly mean is ready within one record interval of the hour's end, at most 15 min on TTN [B7]. R8 is met on paper.

■ C. NO2 error budget (R2, R3, R4)

- **Units and resolution.** 1 ppb = 1.880 µg/m³ at 25 °C; one ADC step of 62.5 µV is 0.25 ppb at the assumed gain [C1].
- **Hourly error.** The hourly terms (1σ) are noise 0.46 ppb after averaging 120 samples, calibration residual 4.39 ppb, drift 0.87 ppb and transfer bias 2.00 ppb, 4.92 ppb in all [C2]. That is a mean absolute error of 3.9 ppb (7.4 µg/m³) against the 5 ppb of R2 [C3]. R2 is at risk: every term except noise rests on literature or an assumption, and only collocation can show it.
- **Annual mean.** Random terms average out over a year, but the transfer bias, the drift residual and the collocation intercept (0.59 ppb from 14 days) do not: 2.26 ppb, or 4.25 µg/m³, against 2 µg/m³ [C4]. Meeting 2 µg/m³ would need the transfer bias held to 0.16 ppb, which no field calibration can promise [C6]. R3 is therefore withdrawn as a requirement and kept as a research question (DDR-002).
- **EU limit.** At 90 % one-sided confidence, a street's annual mean can be classified against 40 µg/m³ only if it is below 34.6 or above 45.4 µg/m³ [C5]. R4 is at risk: it holds for streets well clear of the limit but not for those near it.

■ D. Radiation shield (R5)

- **Plate temperatures.** In 1,000 W/m² sun the cap runs 8.3 K and the bottom plate 1.7 K above ambient; the mean radiant excess seen by the probe is 1.30 K [D2].
- **Probe error.** With inside air at 30 % of a 1 m/s wind the probe reads 0.26 K high from radiation and 0.17 K from warmed air, 0.43 K in all; at 10 % it reads 0.38 K and 0.52 K, 0.89 K in all [D1]. Both are under the 1 °C of R5, but the result depends on the inner air speed, which is assumed.
- **Humidity.** A 0.89 K warm error reads 2.5 % RH low at 50 % RH and 4.2 % RH low at 85 % RH, at 30 °C [D3]. The SHT45 alone meets ± 0.5 °C and ± 3 % RH, but the shield error at high humidity does not. R5 is at risk.

■ E. PM2.5 error with humidity (R1)

- **RH input.** The humidity correction uses RH with an uncertainty of 4.4 % (1 % sensor and the shield error at 85 % RH) [E1].
- **Error against the bound.** Table 2 compares the combined error with the R1 bound of ± 5 µg/m³ or ± 30 % [E2].

Table 2. Hourly PM2.5 error after humidity correction, µg/m³ (bound in parentheses).

RH	10 µg/m ³	25 µg/m ³	50 µg/m ³
50 %	2.2 (5.0)	3.2 (7.5)	5.4 (15.0)
70 %	2.6 (5.0)	4.7 (7.5)	8.7 (15.0)
80 %	3.1 (5.0)	6.3 (7.5)	12.2 (15.0)
85 %	3.6 (5.0)	7.9 (7.5)	15.4 (15.0)
90 %	4.8 (5.0)	11.0 (7.5)	21.7 (15.0)

- **Result.** R1 is met on paper up to 80 % RH at every level tested, which is also the top of the SPS30's recommended humidity range [E3]. Above 85 % RH the particles' water uptake is steep and the correction is uncertain. R1 is at risk for humid climates and nights.

■ F. Geometry, area and mass (R11, R13)

- **Fit.** Inlet plane 3,000 mm above the sidewalk, inside the EU 1,500 to 4,000 mm range; 140° V-saddles and bands cut to length fit 80 to 200 mm poles; no drilling [F1]. R11 is met by design.
- **Layout.** FieldNode's whip clears the pod's drip lid by 15 mm (23 mm in v0.2, before FieldNode's antenna moved 28 mm toward the pod and the pod moved 20 mm away), the clamps are 375 mm apart, the node is 706 mm tall and it stands 173 mm out from the pole face [F2].
- **Area.** Rasterizing the model gives 0.143 m² seen from the street and 0.055 m² along it, against 0.15 m² [F3] (0.126 m² in v0.2). FieldNode's constructable bracket, the adapter plates and the cross arm add to the outline. R13's area limit is met on paper.
- **Mass.** FieldNode core as used 1.89 kg (FieldNode's base node of 2.45 kg less its back plate, V-blocks and bands, 0.56 kg); rail 0.24 kg; adapter plates 0.27 kg; cross arm 0.19 kg; V-saddles 0.18 kg (printed solid; less at 40 % infill); pod shell and floor 0.37 kg; shield plates, cap and spacers 0.27 kg; bands 0.10 kg; shield rods 0.06 kg; sensors, front end, probe, cables, glands, inserts and fixings 0.29 kg [F4]. In all 3.84 kg on the pole against 3.5

kg, and 4.00 kg with FieldNode's sun shield at a hot site; the sensor head and mount are 1.95 kg [F5]. **R13 is not met** on mass.

- **Why it rose.** FieldNode's constructable core is 0.15 kg heavier as AirStreet uses it, and the parts that make AirStreet buildable (adapter plates in place of adapter bars, the cross arm in place of the short shield arm, larger saddles, the separate pod floor, spacers and fixings) add the rest [F6]. Options are in AST-DDR-003, A1.
- **Bands.** Each band runs 487 mm round the 140 mm design pole and its saddle, 326 mm round an 80 mm pole and 664 mm round a 200 mm pole, plus about 100 mm for the housing and tail [F7].

■ G. Wind and mounting

- **Load.** A 35 m/s gust gives 750 Pa and 128 N on the node at about 3.33 m, or 428 N·m at the pole base [G1]. The pole's capacity is for its owner to confirm.
- **Clamps.** The downward load of 78 N (weight and wind on the tilted panel) compares with 1,885 N of friction from two bands, a factor of 24 [G2]. Twist about the pole from wind on the offset shield is 2.42 N·m against 132 N·m [G5]. Both rest on the assumed 1,000 N band preload.
- **Cross arm.** The 30 x 30 x 3 mm angle carries the shield (0.34 kg, 12.1 N of wind) 200 mm to the right of the rail and the pod (0.46 kg) 90 mm to the left: 0.67 N·m vertical and 2.42 N·m horizontal at the rail, 4.5 MPa [G3]. Its two M5 bolts each take about 67 N in the gust [G4].
- **Rail and pod.** The rail at the lower clamp carries 26.6 N and 2.73 N·m, 16.4 MPa in the 40 x 5 mm bar [G6]. Each of the pod's two hanging screws takes about 11 N in the gust, far below what an M5 heat-set insert in ASA holds; the insert pull-out is to be checked at TRL 4 [G7]. Every stress is far below the roughly 150 MPa yield of 6063 aluminium.

■ H. Service and environment (R12, R16)

- **Service time.** Under DDR-002 the pod is exchanged whole: ladder 5 min, unplug two M12 plugs and the probe lead 1.5 min, release the pod 1.5 min, fit the pre-located pod 2 min, reconnect and check glands 1.5 min, uplink check 2 min: about 14 min against 15 min [H1]. The exchange pod arrives with its collocation and CalRig record, and the returned pod is serviced and re-located at the reference site [H2]. R16 is met on paper, with a small margin.
- **Temperature range.** FieldNode is rated for -20 to +45 °C ambient and reaches 58.7 to 73.3 °C inside at 45 °C, while AirStreet's R12 asks for -10 to 50 °C [H3]. The SPS30's recommended range is 10 to 40 °C and 20 to 80 % RH, with absolute limits of -10 to 60 °C and 0 to 95 % RH [H4]. Under DDR-002 FieldNode's sun shield is required at sites above 45 °C, so R12 asks for it there; FieldNode has now designed it (FND-DDR-003), and it fixes to AirStreet's adapter plates (AST-DDR-003). R12 stays at risk; condensation and insects in the pod remain unverified.

■ I. Cost (R15)

- **Totals.** The BOM has 11 lines, all priced: \$434.00 per node, of which the FieldNode core is \$131.00 (FieldNode's \$139.00 less its unused \$8.00 pole mounting kit; costed in FieldNode, AST-DDR-001 D1) and the sensor head \$303.00 [I1].
- **Against the budget.** The sensor head is \$18.00 over the \$285 `budget_usd` [I2]. **R15 is not met.** The rise from \$285.00 is in line 3 (band stock, adapter plates and cross arm, +\$4), line 8 (spacers, +\$1), line 9 (probe tube, +

\$1), line 10 (probe lead with its M8 plug, +\$4) and line 11 (inserts, glands, probe socket, terminal block and fixings, +\$8). Options are in AST-DDR-003, A2.

- **Running costs.** A replacement NO2 sensor about every 2 years and the collocation time are not in the BOM [I3].

■ J. Results

Table 3. Requirement status (AST-REQ-001 v0.5). Counts: 2 not met, 5 at risk, 4 met on paper, 4 met by design, 1 withdrawn [J1].

ID	Requirement	Value	Target	Status
R13	Light and compact	3.84 kg (4.00 kg with the sun shield); 0.143 m ² [F3, F5]	3.5 kg; 0.15 m ²	Not met
R15	Low cost	Sensor head \$303.00; \$434.00 with FieldNode [I1]	Sensor head \$285 (budget_usd)	Not met
R1	PM2.5	Within the bound up to 80 % RH, outside it above 85 % RH [E2]	±5 µg/m ³ or ±30 % hourly	At risk
R2	NO2	Hourly MAE about 3.9 ppb [C3]	5 ppb or less	At risk
R4	EU limit classification	Decisive outside 34.6 to 45.4 µg/m ³ [C5]	90 % confidence at 40 µg/m ³	At risk
R5	T and RH	Shield 0.43 to 0.89 K at 1 m/s; 4.2 % RH low at 85 % RH [D1, D3]	±0.5 °C, ±3 % RH; radiation 1 °C or less	At risk
R12	Survive outdoors	FieldNode sun shield required above 45 °C, designed and fits the adapter plates; SPS30 recommended to 80 % RH [H3, H4]	-10 to 50 °C; shield above 45 °C	At risk
R8	Reporting	5 min on the private gateway; 15 min on TTN, fair use met up to SF9; 7 days = 40.3 kB [B6, B8]	5 min private, 15 min TTN; hourly within 15 min; 7 days	Met on paper
R9	Power	87.0 mW design load; 2.33 Wh/day against 7.75 Wh [A3, A7]	115 mW or less; energy neutral	Met on paper
R10	Duty cycle	0.082 % at SF9, 0.60 % at SF12 [B3]	Below 1 %	Met on paper
R16	Serviceable	About 14 min for a pod exchange [H1]	Pre-located pod in 15 min	Met on paper
R6	Stated calibration	CalRig for T, RH, PM; collocation per D3	Calibration record per node	Met by design
R7	Raw data	Raw WE, AE, PM, T, RH and version tag	Raw signals kept	Met by design
R11	Pole mounting	80 to 200 mm, V-saddles and bands, inlet 3.0 m [F1]	No drilling; 1.5 to 4 m	Met by design
R14	Levels only	Levels, T and RH only	No images, audio or identifiers	Met by design
R3	NO2 near the WHO guideline	4.2 µg/m ³ annual (1σ) [C4]	Withdrawn; research question	Withdrawn

Table 4. TRL 2 numbers checked (v0.1).

TRL 2 figure	TRL 3 value	Action
Sensor power about 41 mW, 45 mW with margin	35.0 mW typical, 48.0 mW design	Precis updated
Airtime about 1.3 s at SF12, 0.5 %	1.81 s, 0.60 %	Precis and requirements updated
Mass about 3.0 kg	3.65 kg (FieldNode 2.41 kg, not 1.7 kg)	R13 not met in v0.1; see Table 5
Frontal area about 0.09 m ²	0.138 m ² from the model	R13 not met in v0.1; see Table 5
Cost \$391 per node; \$265 sensor head	\$409.00; \$283.00	BOM repriced
NO2 hourly error about 5 ppb	About 3.9 ppb MAE, budget of assumed terms	R2 stays at risk

Table 5. Changes from v0.1 under AST-DDR-002.

Quantity	v0.1	v0.2	Decision
PM run per record	30 s	60 s	PM run length
Design sensor load	48.0 mW	87.0 mW	PM run length
Autonomy without sun	11.9 days	6.6 days	PM run length
Reporting	5 min on any network	5 min private gateway; 15 min on TTN	Fair use on TTN
Mass on the pole	3.65 kg	3.26 kg	Back plate left off
Frontal area	0.138 m ² against 0.12 m ²	0.126 m ² against 0.15 m ²	Back plate left off; area limit relaxed
Sensor head cost	\$283.00 against \$180	\$285.00 against \$285	Budget figure; adapter bars
Service	14 min sensor swap, relocation needed	14 min pod exchange	Pod as exchange unit
R3	Not met	Withdrawn, research question	WHO guideline

Table 6. Changes from v0.2 under AST-DDR-003 (design for construction).

Quantity	v0.2	v0.3	Reason
Mass on the pole	3.26 kg	3.84 kg (4.00 kg with the sun shield)	FieldNode's constructable core; adapter plates, cross arm, saddles, pod floor, fixings
Frontal area	0.126 m ²	0.143 m ²	FieldNode's bracket, plates and cross arm
Wind on the node	114 N, 379 N·m	128 N, 428 N·m	Larger area
Sensor head cost	\$285.00	\$303.00	Parts added for construction
FieldNode core cost	\$126.00	\$131.00	FieldNode's constructable BOM less its pole mounting kit
Whip clearance to the pod	23 mm	15 mm	FieldNode's antenna moved; pod moved 20 mm left
Shield arm	20 x 8 mm bar, 4.6 MPa	30 x 30 x 3 mm cross arm carrying pod and shield, 4.5 MPa	Pod and shield needed fixings
R13 and R15	Met on paper	Not met	Proposed to Amish, AST-DDR-003 A1 and A2